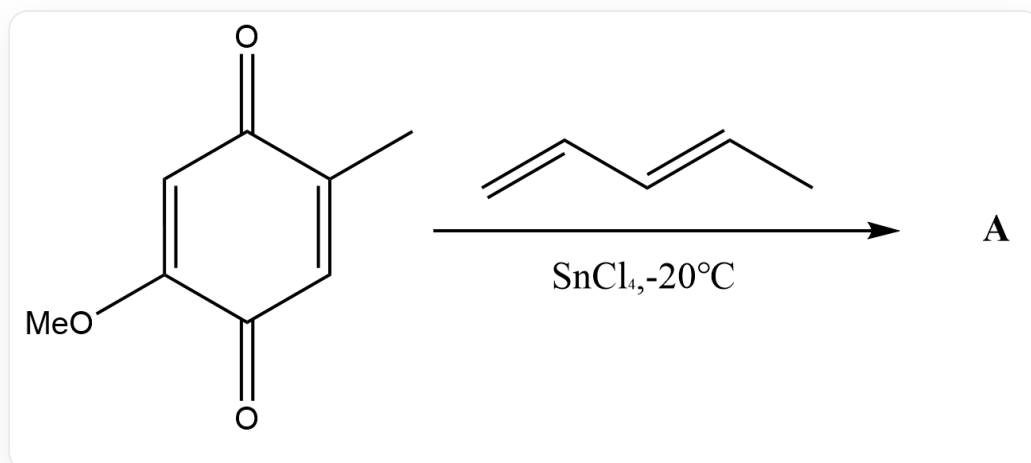
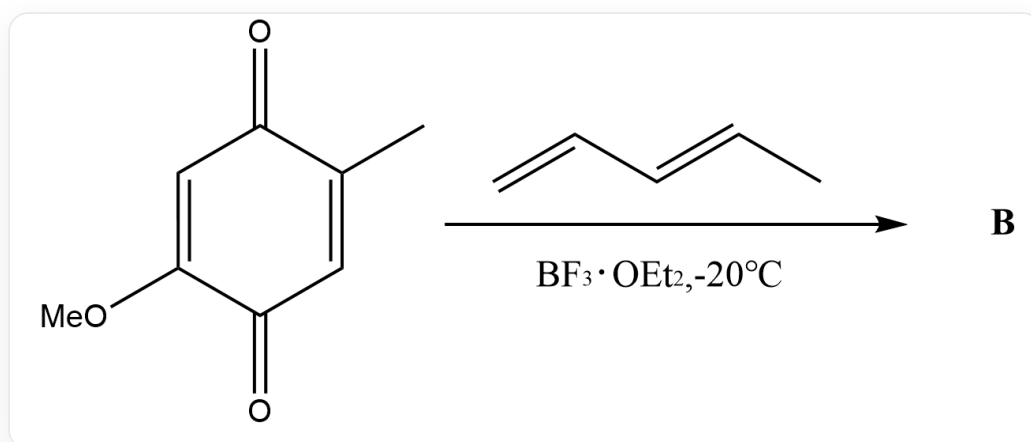


Question



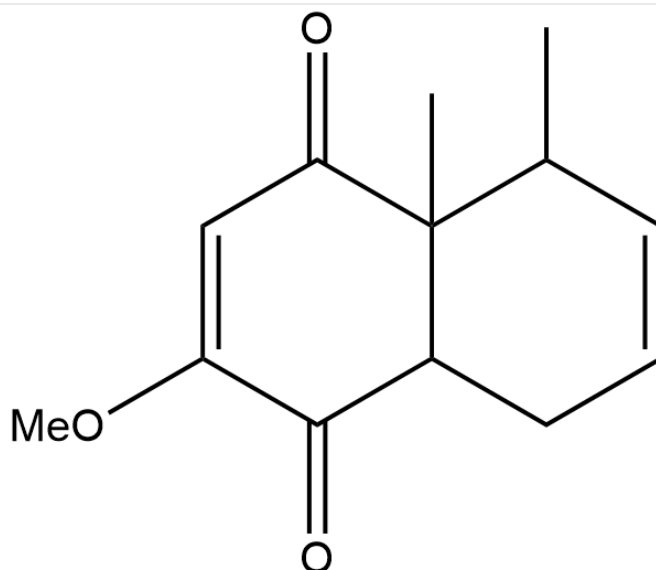
O=C(C=C1OC)C(C)=CC1=O>C=C/C=C/C.Cl[Sn](Cl)(Cl)Cl>[A], the reaction was carried out at -20°C



O=C(C=C1OC)C(C)=CC1=O>C=C/C=C/C.FB(F)(O(CC)CC)F>[B], the reaction was carried out at -20°C

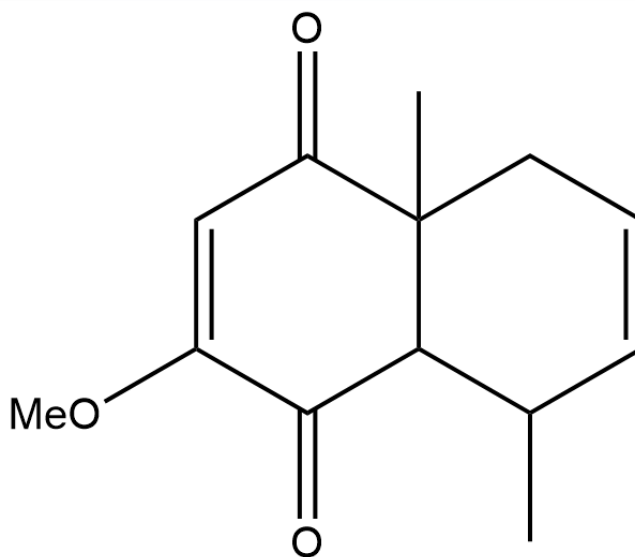
Please provide the most likely products A and B for the above two reactions, respectively (without considering stereochemistry)

A.



O=C(C=C1OC)C(C(C)C)=CC2(C)C2C1=O, Product A

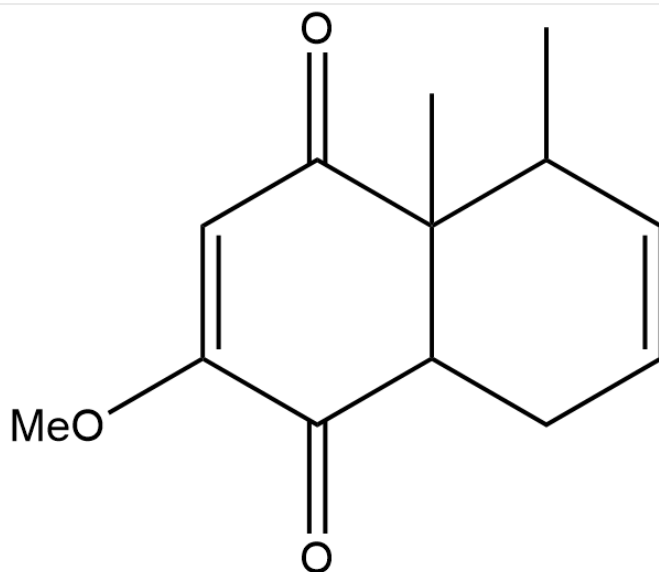
Product A



O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O, Product B

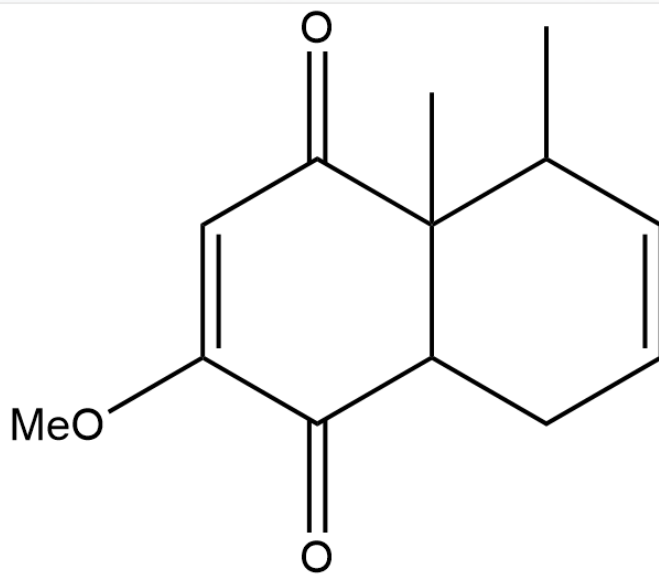
Product B

B.



O=C(C=C1OC)C(C(C)C)=CC2(C)C2C1=O, Product A

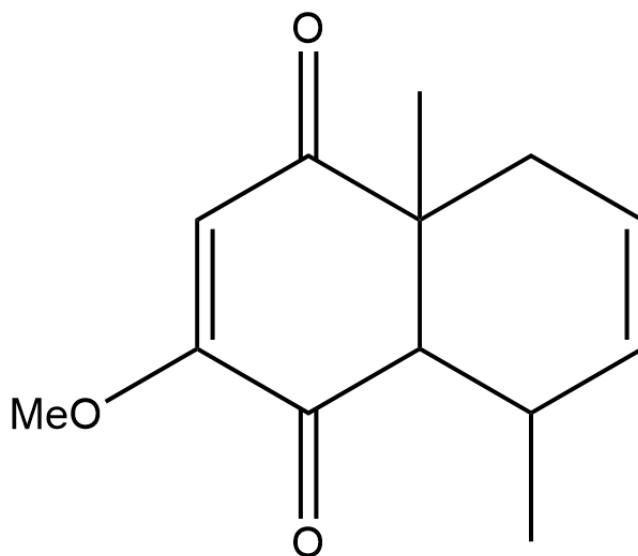
Product A



O=C(C=C1OC)C(C(C)C)=CC2(C)C2C1=O, Product B

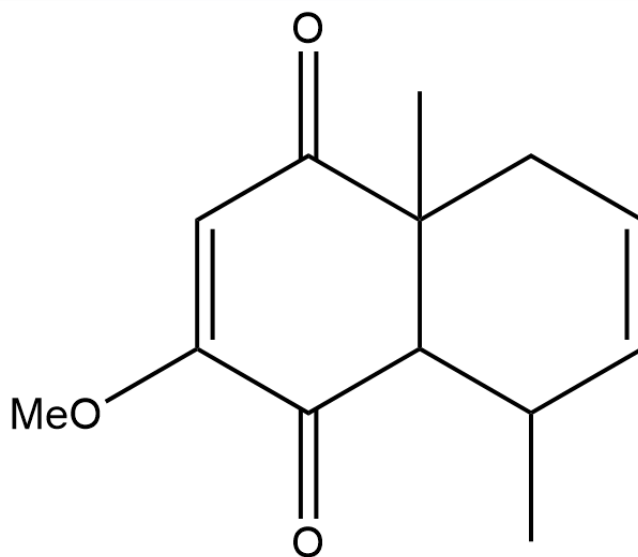
Product B

C.



O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O, Product A

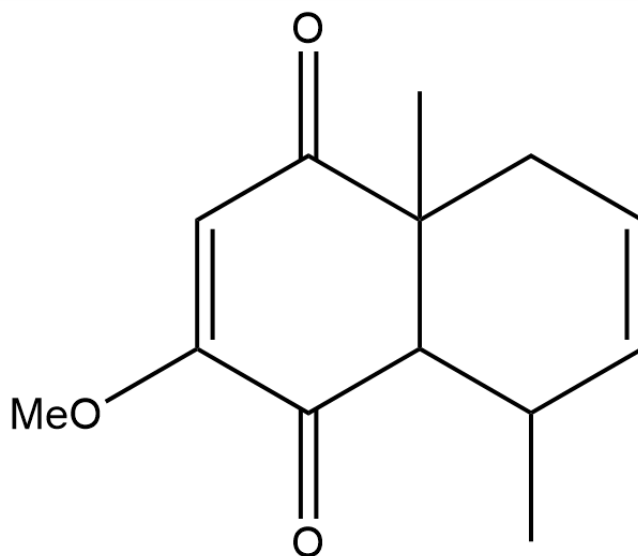
Product A



O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O, Product B

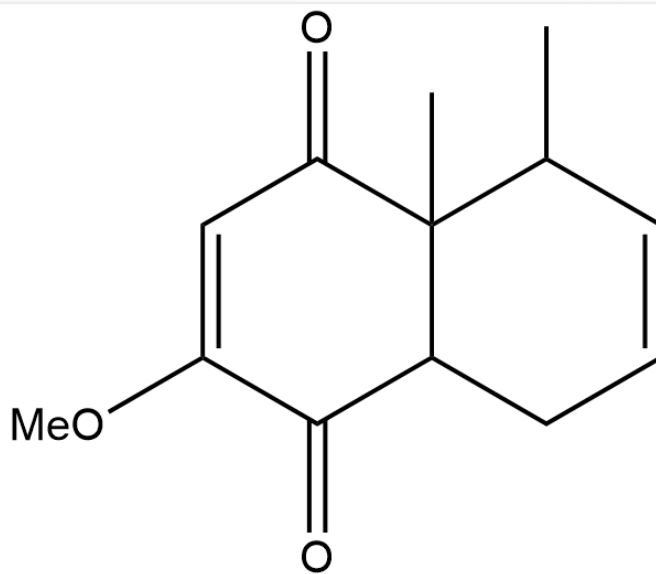
Product B

D.



O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O, Product A

Product A



O=C(C=C1OC)C(C(C)C=CC2)(C)C2C1=O, Product B

Product B

Answer

Correct Answer: **D**

Detailed Explanation

SnCl_4 can simultaneously coordinate with both the carbonyl and methoxy groups, thereby activating the lower carbonyl.

CHECKPOINT

1 PTS

SnCl_4 can simultaneously coordinate with both the carbonyl and methoxy groups, thereby activating the lower carbonyl

In contrast, BF_3 can only form tetracoordination and preferentially coordinates with the electron-rich upper carbonyl, activating the upper carbonyl.

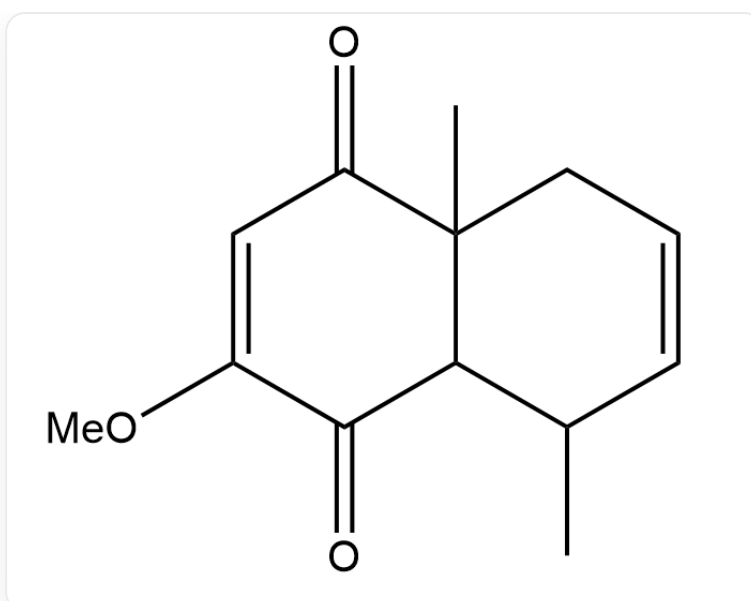
CHECKPOINT

1 PTS

In contrast, BF_3 can only form tetracoordination and preferentially coordinates with the electron-rich upper carbonyl, activating the upper carbonyl

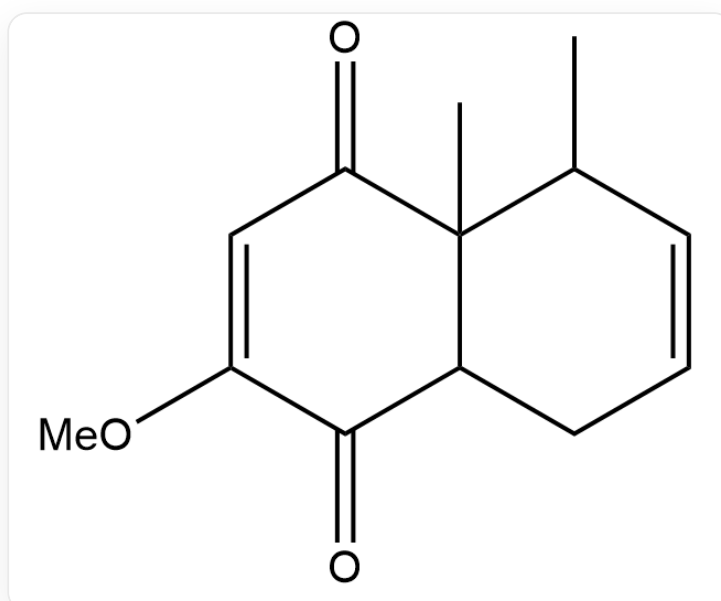
For the diene component, the alkyl group is electron-donating, so the side connected to the alkyl group becomes positively charged, while the other end becomes negatively charged. Reacting with the corresponding sites of the dienophile yields products A and B:

A.



O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O, Product A

B.



O=C(C=C1OC)C(C(C)C=CC2)(C)C2C1=O, Product B

Therefore, option D is correct.

CHECKPOINT

1 PTS

Product A is O=C(C=C1OC)C(CC=CC2C)(C)C2C1=O

CHECKPOINT

1 PTS

Product B is O=C(C=C1OC)C(C(C)C=CC2)(C)C2C1=O